Current makes a magnetic field

antocciphysics

UNIT 7 / DAY 13 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect current direction with the magnetic field around a wire.

SUCCESS CRITERIA - I can...

1. Sketch circular field lines around a straight wire.
2. Use a stated right-hand rule.
3. Predict the effect of reversing current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **6 Ω** resistor is in series with two parallel **8 Ω** resistors across **20 V**. **Find parallel equivalent**, total resistance and supply current.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $B = \mu_0 I/(2\pi r)$; $\mu_0 = 4\pi \times 10^{-7}$ **T·m/A**.
Long straight wire. Right thumb follows conventional current; curled fingers show field direction.

Worked reasoning: Around a straight wire carrying conventional current out of the page, the magnetic field circles counterclockwise by the right-hand rule. A dot means out of page.

Your turn: Draw three field arrows around the wire and **explain what the dot represents**.

Current out of page (dot). Add field arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Conventional current points out of the page. Use the right-hand rule to **draw the surrounding magnetic field** direction.

Current out of page (dot). Add field arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A long straight wire carries **5.0 A** out of the page. **Calculate B** at **0.020 m** using $B = \mu_0 I/(2\pi r)$, $\mu_0 = 4\pi \times 10^{-7}$ **T·m/A**. **Mark the location** and local field direction on the diagram.

Current out of page (dot). Add field arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

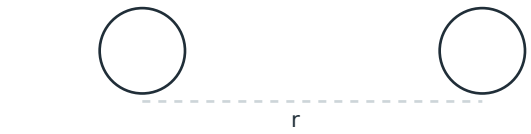
05 Explain your reasoning

What happens to field direction if the current reverses while its magnitude stays the same?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A neutral object is attracted to an object carrying **+4 nC**.
Explain how attraction can occur without giving the neutral object a net charge.
Label charges and both interaction forces.



Charge A Charge B

Marzano check - circle: 1 2 3

07 Exit check

Current triples and distance doubles. Express new B as a multiple of the old value.

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

For a long straight wire use $B=\mu_0 I/(2\pi r)$, with $\mu_0=4\pi\times10^{-7}$ **T·m/A**. **Find B** for **5 A** at **0.10 m**; then double distance.

Cross = current into page. Add magnetic-field arrows.



Marzano check - circle: 1 2 3